

Kyle Sabado

+1 647-673-5311 | kyle.sabado@mail.utoronto.ca | linkedin.com/in/kylesabado
github.com/Kylx28 | Portfolio: kylx28.github.io/

EDUCATION

University of Toronto

Sept. 2022 – May 2027

Bachelor of Applied Science in Computer Engineering, Minor in Robotics

Toronto, ON

Relevant Courses: Matrix Algebra & Optimization, Probabilistic Reasoning, Real Analysis, Data Structures & Algorithms, Operating Systems, Machine Learning, Deep Learning, Control Systems, Robot Modeling & Control

TECHNICAL SKILLS

Languages: Python, C++, C, MATLAB, Java, Bash, Verilog

ML & Distributed Systems: PyTorch, Hugging Face, Weights & Biases, MuJoCo, vLLM, NCCL, Ray

Tools & Frameworks: Linux, Git, Docker, ROS/ROS 2, OpenCV, Google Cloud

EXPERIENCE

Reinforcement Learning Research Intern

May 2026 – Present

MARMoT Lab (National University of Singapore)

Singapore

- Developing a **PPO-based reinforcement learning** policy with expert action priors to improve planner trajectories for continuous-space **multi-agent pathfinding**.
- Built a planning pipeline using **NVIDIA SWAGGER** for sparse graph generation and Safe Interval Path Planning (SIPP) to generate multi-agent trajectories.

AI Networking Intern

May 2025 – Apr. 2026

Huawei Canada

Toronto, ON

- Improved **Mixture-of-Experts (MoE) LLM inference latency by 20%** by designing a topology-aware expert placement system in **Python**, using expert sharding and replication to reduce communication overhead.
- Developed a distributed **collective communication scheduler** for multi-tenant GPU workloads using **Python** and TCP/RPC, improving training throughput by up to **30%** under high-contention scenarios.
- Trained and profiled LLMs across **16 GPUs** using **Megatron-LM**, **PyTorch FSDP**, and **PyTorch Profiler**, evaluating tensor, expert, and data parallelism to identify performance bottlenecks.
- Implemented **AlltoAllv** using **NCCL** primitives in **vLLM**, enabling variable-size data transfers for distributed MoE inference.

DESIGN TEAMS

Motion Prediction Team Member

Sept. 2026 – Present

aUToronto

Toronto, ON

- Implemented a **ROS 2** motion prediction pipeline in **C++**, integrating tracked objects and map data with the learned **SIMPL** model to predict and visualize vehicle and pedestrian trajectories.

Localization Team Member

Sept. 2023 – Dec. 2025

UofT Autonomous Drone Racing

Toronto, ON

- Implemented an **EKF-based visual-inertial odometry** pipeline in **C++/ROS**, fusing IMU and camera data with **FAST/ORB** feature tracking for real-time localization.

PROJECTS

Learned Humanoid Robot Get-Up | *MuJoCo, PyTorch*

Sept. 2026 – Present

- Trained a **Unitree G1** policy in **MuJoCo** that achieves sustained standing from a supine position on rigid ground using **PPO**, **residual reinforcement learning**, and expert demonstrations for guided exploration.

Deep Learning Skin Cancer Classifier | *PyTorch, Scikit-learn*

May 2025 – Aug. 2025

- Trained an **ensemble CNN** with an **expert router**, transfer learning, data augmentation, and class rebalancing, achieving **80% balanced accuracy** on skin cancer classification and outperforming individual models by **5%**.

PUBLICATIONS

M. Ammous*, **K. Sabado***, M. Saif and S. Valaee, "Cooperative Localization and Tracking Using RISs and Sidelink Communications," 2025 IEEE Wireless Communications and Networking Conference (WCNC), Milan, Italy, 2025.